UNIVERSITY OF CAMBRIDGE

NST1
NATURAL SCIENCES TRIPOS Part IB

Friday 13 June 2025 13.30 to 16.30 (*3 hours plus any pre-agreed individual adjustment*)

Paper PHA/2

PHARMACOLOGY WRITTEN PRACTICAL PAPER

Answer **all four** questions. Each **question** is worth **20 marks in total**.

Extended Answer questions (2) and (4) have no word limit

Diagrams or other handwritten material
Where necessary to perform, show your mathematical workings.
Where appropriate (e.g., for formulae, equations, diagrams or mathematical calculations) you may hand write your answers/workings using the Inspera Scan Paper provided. You should not copy and paste images from any other documents.

Please Note

- Inspera Scan question codes MUST be noted during the exam.
- Please ensure that each Inspera Scan page is labelled appropriately, including your Blind Grade Number.
- Inspera Scan pages MUST be placed in the Inspera Scan folder at the completion of the exam and left on your desk.
- Inspera scan pages will be scanned by administrators after the completion of the exam.

Stationery requirements

- Rough work pad
- Inspera Scan pages
- Inspera Scan Folder
- Equation Sheet
- Figures and Tables Sheet

Calculator – students are permitted to use an approved calculator.

An equation sheet and sheet with figures and tables is included and may be consulted during the exam.

Question 1

The Epidermal Growth Factor Receptor (EGFR) is a transmembrane protein whose function is dysregulated in several forms of cancer, driving cell proliferation and tumour growth. **EB45** is a first-generation EGFR blocker and is currently approved for the treatment of late-stage non-small cell lung cancer (NSCLC). In the race to develop the second-generation EGFR blockers, a start-up biotech developed **GU46** which is an experimental drug undergoing preclinical evaluation.

To determine if **GU46** binds to EGFR, cell membranes from NSCLC cells were incubated with varying concentrations of **^{3}H-GU46** (specific activity: 2000 Ci/mmol, 1 Ci = 1.2 x 10^{12} dpm) in a final reaction volume of 2 mL, containing cell membranes at a protein concentration of 0.05 mg/mL. After a 40 minute incubation, the reaction was stopped, and the membranes were washed with ice-cold buffer. Subsequently the radioactivity was assessed using a scintillation counter. The same experiment was then repeated, but with 0.1 mM unlabelled **GU46** added to the reaction. The results of both experiments are shown in **Table 1** below. Specific binding was calculated and specific binding / [^{3}H-GU46] was then plotted against the specific binding to produce the Scatchard plot in **Figure 1.**

Table 1. Binding of ^{3}H-GU46 to NSCLC cell membranes.

[^{3}H-GU46] (nM)	**Binding of ^{3}H-GU46 in the absence of 0.1 mM GU46 (dpm)**	**Binding of ^{3}H-GU46 in the presence of 0.1 mM GU46 (dpm)**
100	49	20
500	210	85
1000	410	200
3000	850	440
5000	1250	750
10000	2200	1600

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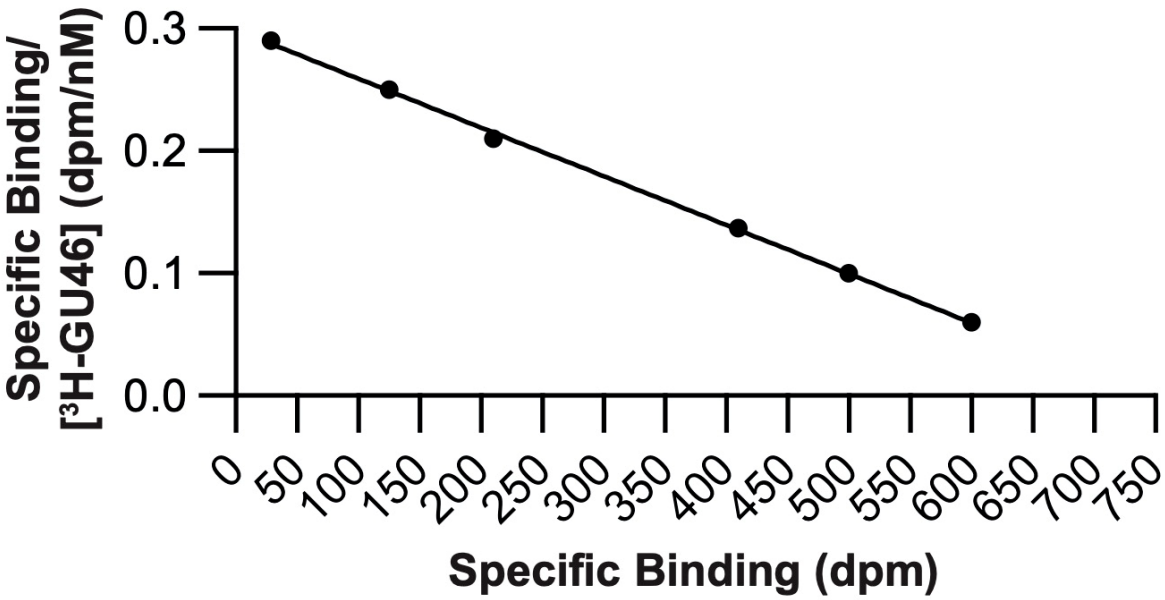

Figure 1. Scatchard plot in which the specific binding / [^{3}H-GU46] (dpm/nM) is plotted against the specific binding of ^{3}H-GU46 (dpm). (Data from Table 1).

A). Why is it important that the cell membranes were washed after incubation with ^{3}H-GU46? And why is it important to use ice-cold buffer? (2 marks)

B). Calculate the association constant (K_A) and dissociation constant (Kd) of the binding of ^{3}H-GU46 to EGFR receptors, as well as the receptor density in NSCLC cell membranes, assuming that one ^{3}H-GU46 binds per receptor. (4 marks)

SCI10 and **PH23** are two experimental drugs developed by two competing pharma companies and are undergoing preclinical evaluation. **SCI10** and **PH23** are supposed to act also on EGFR and the companies want to assess how they compare to **EB45** and **GU46.** The binding affinity of **SCI10**, **PH23** and **EB45** was compared to that of **GU46** in radioligand competition binding assays. Cell membranes from NSCLC cell lines were prepared as described above and were incubated with 1000 nM **^{3}H-GU46** in the presence of different concentrations of the three unlabelled drugs. Results obtained are provided in **Table 2** and plotted in **Figure 2**.

[Question 1 continues on the next page]

Table 2. Inhibition of ^{3}H-GU46 binding to NSCLC cell membranes by SCI10, PH23, or EB45.

[SCI10] (nM)	Binding of ^{3}H-GU46 (dpm)	[PH23] (nM)	Binding of ^{3}H-GU46 (dpm)	[EB45] (nM)	Binding of ^{3}H- GU46 (dpm)
0	349	0	350	0	352
30	341	30	345	30	318
100	321	100	335	100	274
300	276	300	300	300	190
1000	193	1000	260	1000	153
3000	152	3000	200	3000	152
10000	150	10000	155	10000	150

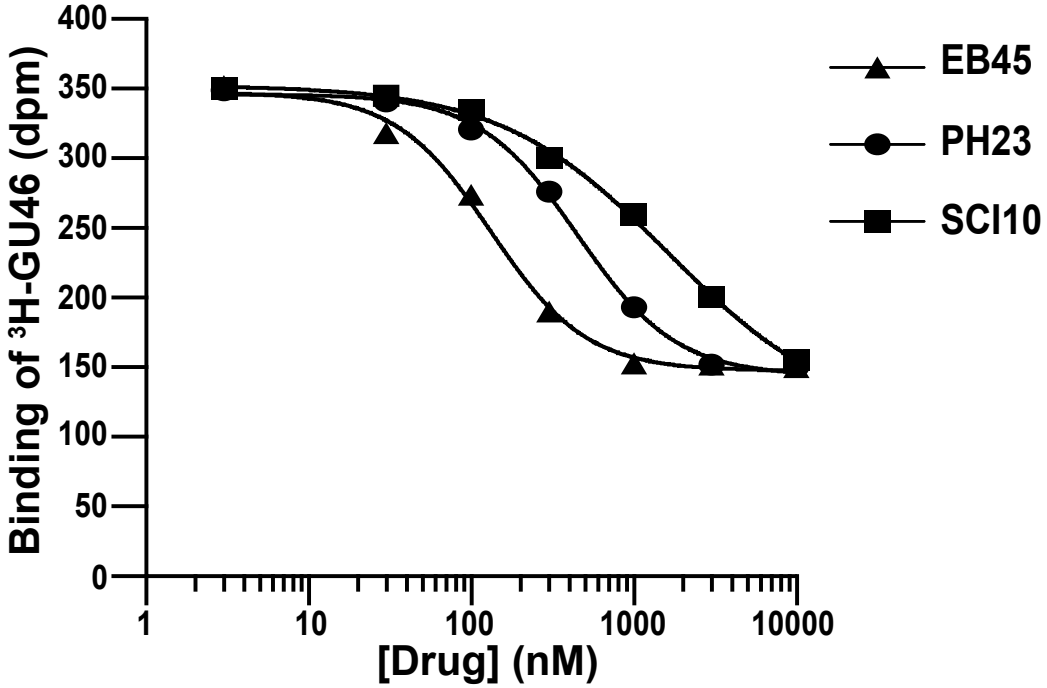

Figure 2. Binding of ^{3}H-GU46 (dpm) to NSCLC cell membranes at increasing concentrations of SCI10, PH23, or EB45 (Data from Table 2).

C). Calculate the binding affinity for SCI10, PH23, and EB45. (5 marks)

D). What can you conclude about the results in Table 2 and Figure 2? (2 marks)

To determine the biological impact of these drugs on cancer cells, the researchers performed two further experiments. First, they incubated equal numbers of NSCLC cells *in vitro* (in tissue culture) with 500 nM of either **SCI10**, **PH23**, **GU46** or **EB45** for 24-hours, and then performed a cell viability assay to quantify the proportion of cells alive. The results of this experiment are plotted in **Figure 3**.

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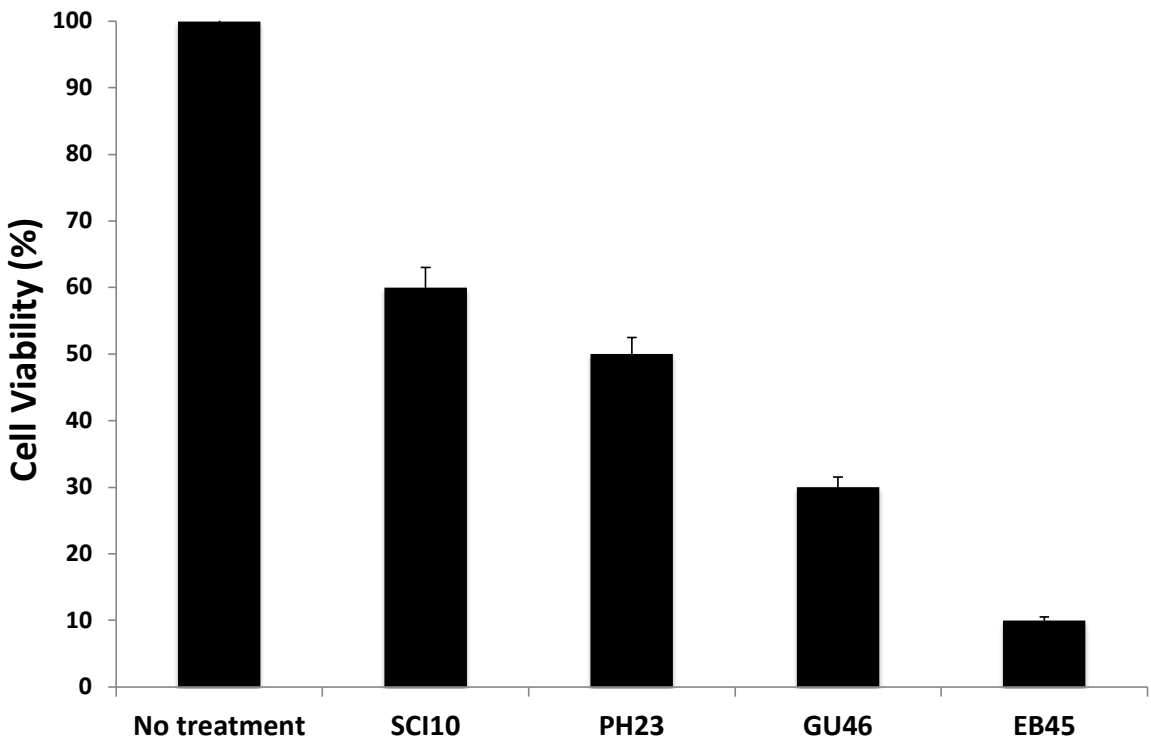

Figure 3. A histogram plot of the % NSCLC cell viability, 24 hours post treatment with 500 nM of either SCI10, PH23, GU46, or EB45 *in vitro*.

E). Comment on the results of the *in vitro* treatment in Figure 3. (2 marks)

The second experiment the researchers performed was *in vivo*. Equal numbers of NSCLC cells were injected subcutaneously in immune-compromised mice. 7 days later, the mice received a 500 nM dose of either **SCI10**, **PH23**, **GU46** or **EB46** once a day for a further 7 days. Tumour volume was measured at the end of the treatment. The results are plotted in **Figure 4.**

[Question 1 continues on the next page]

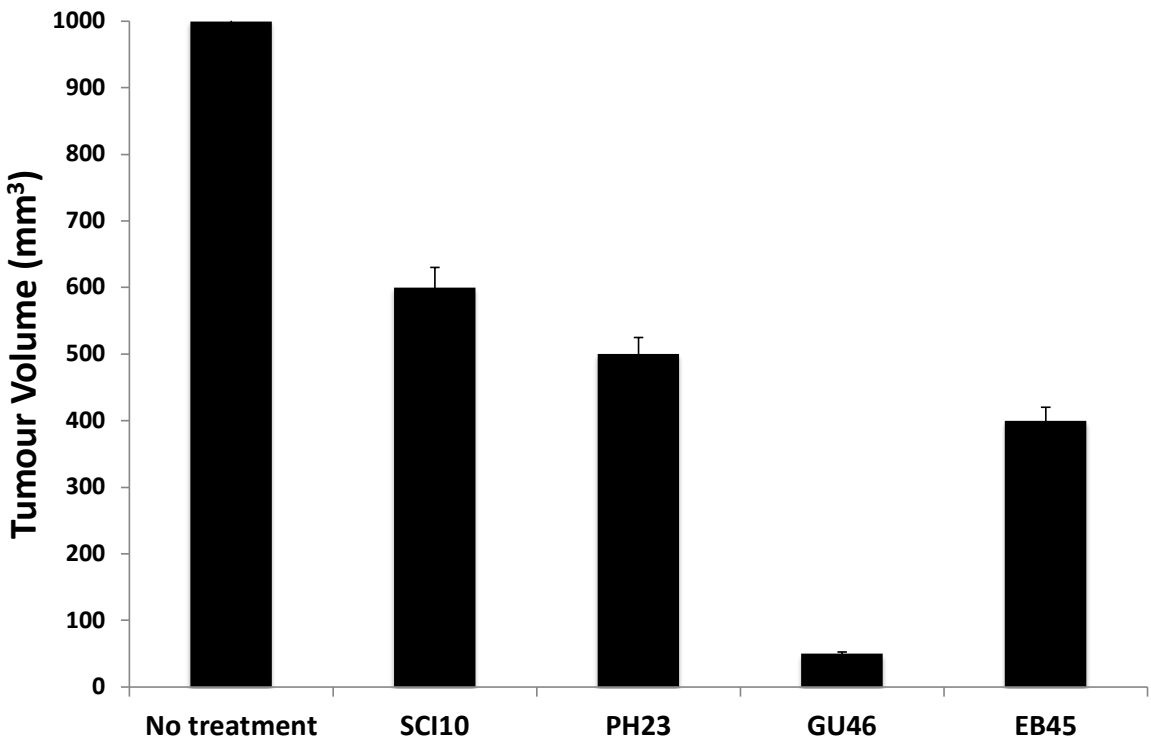

Figure 4. A histogram plot of the tumour volume, 7-days post treatment with 500 nM of either SCI10, PH23, GU46, or EB45 *in vivo*.

F). Comment on the results of the *in vivo* treatment in Figure 4. (2 marks)

G). Discuss why the results of the *in vitro* and *in vivo* experiments do not match. (3 marks)

End of Question 1

[Question 2 starts on the next page]

Question 2

What are key aspects that determine how well and how selectively a drug can control cell proliferation and survival? (20 marks)

End of Question 2

[Question 3 starts on the next page]

Question 3

A novel antibacterial agent (AD2025-P) is currently tested for the treatment of complicated skin and soft tissue infections and community-acquired pneumoniae in adults. In contrast to previous drugs with structural similarity, AD2025-P is also active against some antibiotic-resistant bacteria. AD2025-P is a prodrug that is very rapidly converted in plasma into active AD2025 **(Figure 1)**. AD2025-P does not possess sufficient chemical stability for drug storage in any formulation but, compared to AD2025, has an increased water solubility that enables intravenous administration.

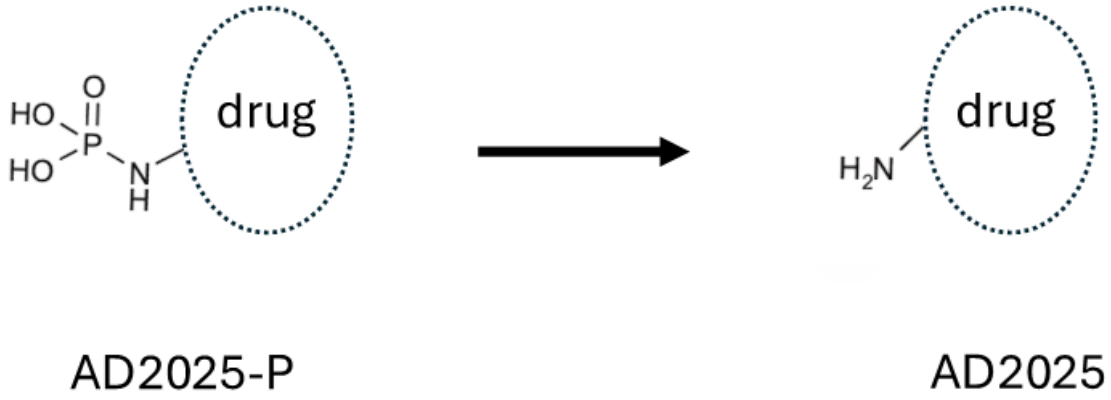

Figure 1. Enzymatic conversion of AD2025-P to AD2025 in plasma.

A) What is the most likely enzyme carrying out this conversion? (1 mark)

To test the impact of different dosing strategies, the researchers investigated the pharmacokinetics of AD2025-P in 20 healthy human volunteers (referred to as the q12h group). A single dose of 600 mg of AD2025-P was administered intravenously over 1 h during which the plasma concentration of the active metabolite, AD2025, was measured. The plasma concentration of AD2025 was then followed for another 11 h **(Figure 2A and 2B, and Table 1).**

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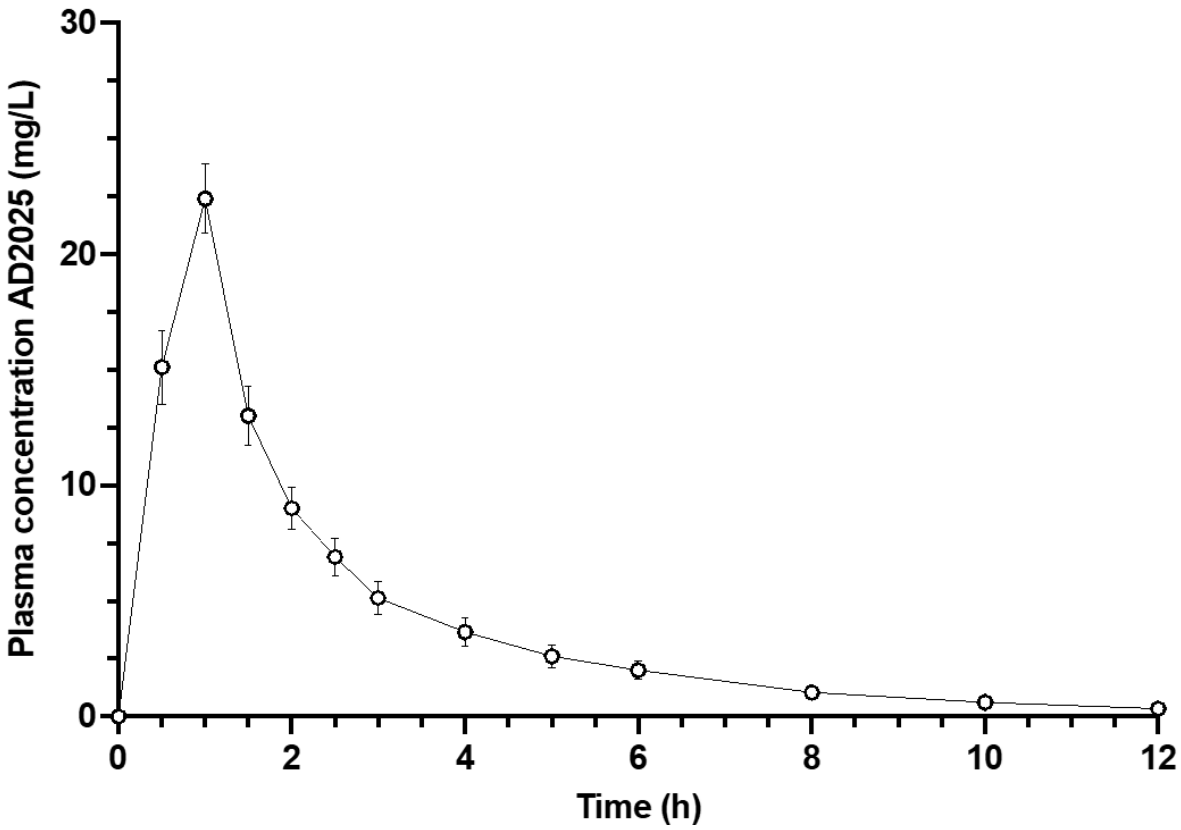

Figure 2A. Plasma concentration of active AD2025 versus time, after the single infusion of 600 mg of prodrug AD2025-P in 1h for the q12h volunteer group. Data are shown as mean ± SD.

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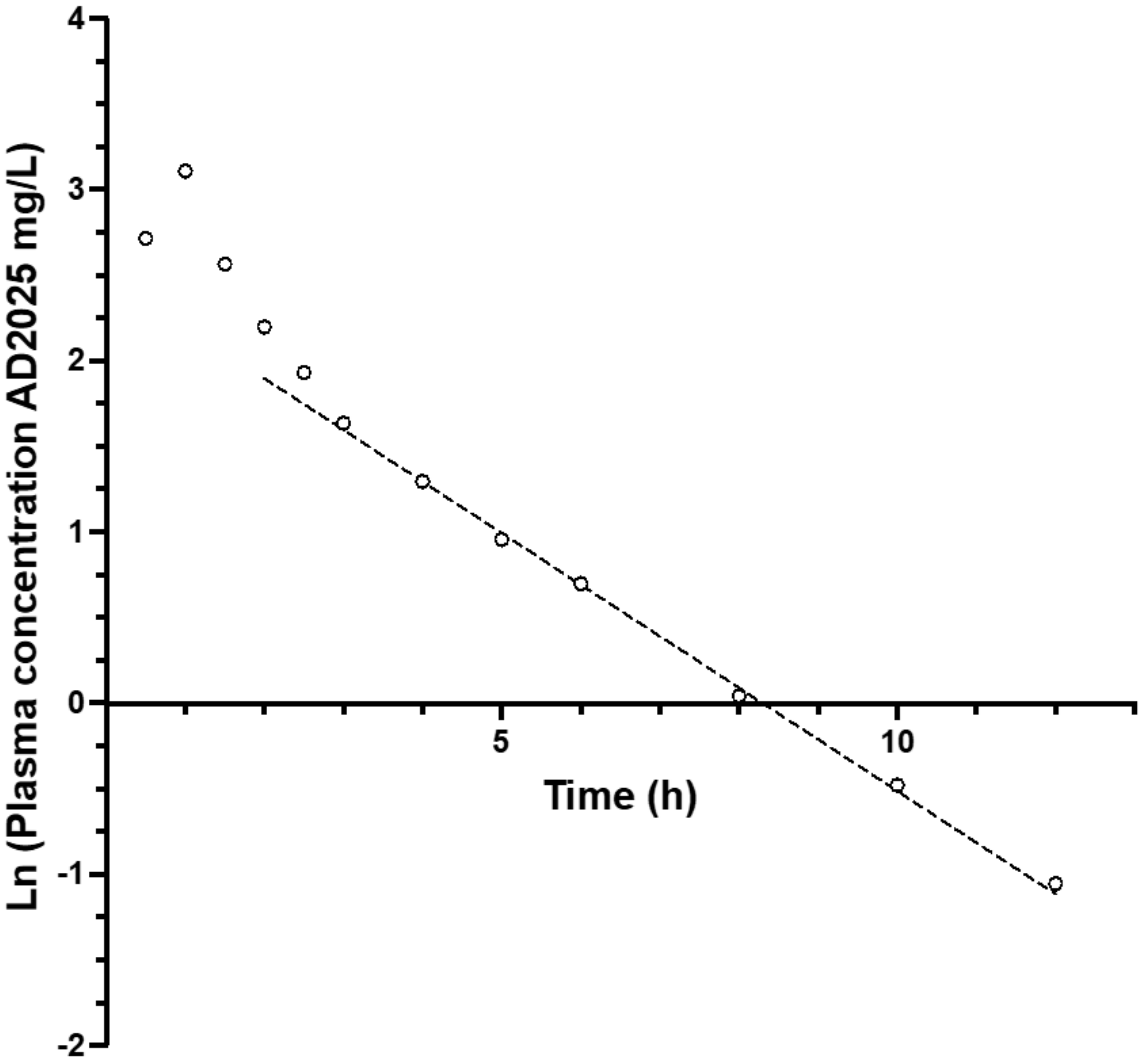

Figure 2B. Data in Figure 2A plotted as the natural log (Ln) of the mean plasma concentration versus time.

[Question 3 continues on the next page]

Table 1. Data plotted in Figure 2A and 2B

Time (h)	Mean plasma concentration AD2025 (mg/L) q12h group	Ln (plasma conc. mg/L) q12h group
0	0.000	
0.5	15.110	2.715
1.0	22.400	3.109
1.5	13.000	2.565
2.0	9.000	2.197
2.5	6.900	1.932
3.0	5.131	1.635
4.0	3.650	1.295
5.0	2.605	0.957
6.0	2.010	0.698
8.0	1.042	0.041
10.0	0.620	-0.478
12.0	0.348	-1.056

B). If 100% of the prodrug in plasma is converted into active AD2025, determine the half-life for elimination of AD2025, and the clearance and volume of distribution in the terminal elimination phase. (7 marks)

C). After the drug infusion step is completed, the terminal elimination phase in Figure 2B is not immediately observed. Considering the calculated pharmacokinetic parameters under B, how do you interpret this phenomenon? (2 marks)

The plasma concentration-time profile of AD2025 was assessed after the administration of repeated intravenous doses of AD2025-P, every 12 hours for the q12h volunteer group (open circles in **Figure 3**). The researchers compared this to a regimen in which the 600 mg dose of AD2025-P was administered intravenously over 2 h for every 8 h in another group of 20 healthy human volunteers (the q8h group) (open squares in **Figure 3**).

[Question 3 continues on the next page]

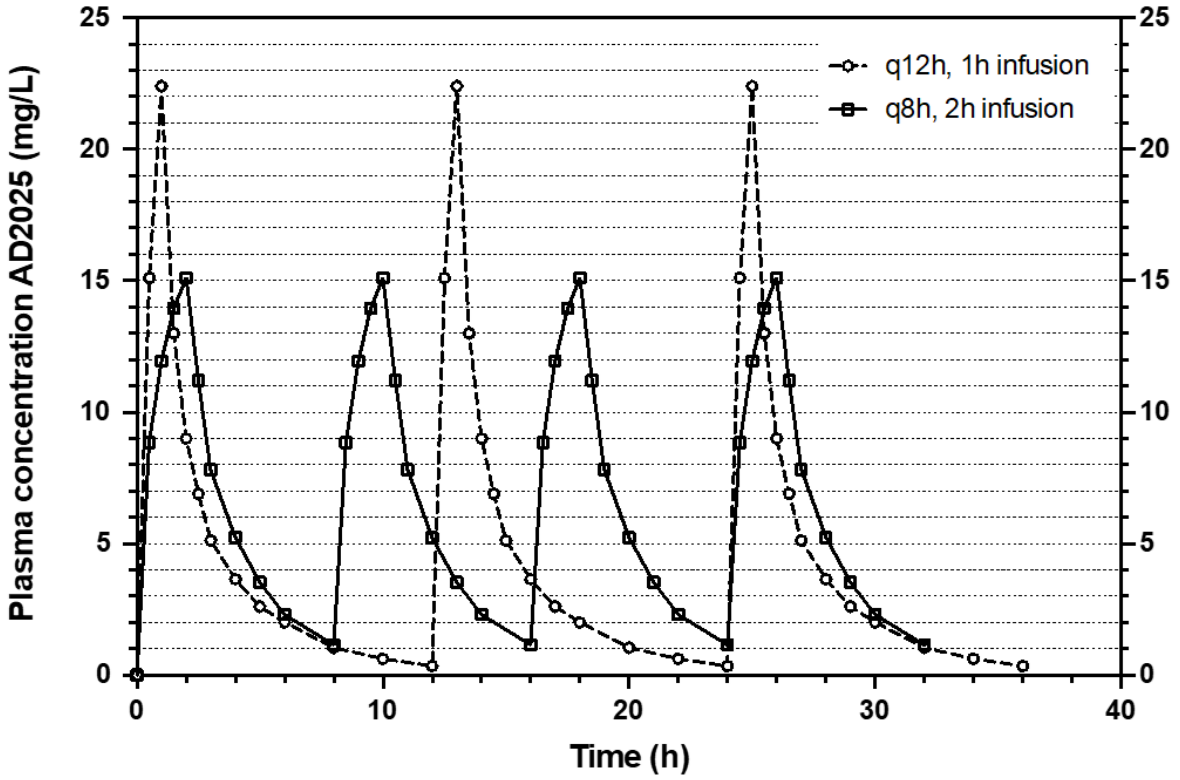

Figure 3. Plasma concentration (mean values only) versus time profile for AD2025 at steady state during the repeated infusions of a 600 mg dose of AD2025-P in 1h for the q12h volunteer group (open circles) and 2h for the q8h volunteer group (open squares). The dose was repeated every 12 hours for the q12h group, and every 8 hours for the q8h group.

D). Using the pharmacokinetic parameters determined in (B) *calculate* the C_{max} and C_{min} that are achieved for the repeated dosing regimen at steady state in the q12h volunteer group (Figure 3). How do these calculated concentrations compare to the experimental values? Please explain. (3 marks)

An important pharmacodynamic parameter predictive of antibiotic efficacy is the percentage of time in each dosing interval during which the concentration of an antimicrobial agent must be equal to, or above, the Minimal Inhibitory Concentration (MIC) of the target bacterium (referred to by the parameter: %*f*T > MIC). For some bacterial pathogens, the MIC and %fT>MIC values for AD2025 are listed in **Table 2**.

Table 2. Data of clinical pharmacodynamic studies using AD2025

Bacterial pathogen	MIC (mg/L)	%fT > MIC
Staphylococcus aureus	1	33
Methicillin-resistant *S. aureus* (MRSA)	2	60
Pseudomonas aeruginosa	5	62
Beta-lactam resistant P. aeruginosa	10	78

[Question 3 continues on the next page]

E). Using Figure 3 and Table 2, determine whether the q8h and q12h dosing regimens for AD2025-P will be effective in treating infections by the listed bacterial pathogens. How do the two dosing regimens compare? (3 marks)

F). How could these pharmacokinetic studies on AD2025-P be extended? (2 marks)

G). What properties of AD2025 could be improved, to facilitate ease of use as an antibacterial agent, and to enhance drug efficacy? (2 marks)

End of Question 3

[Question 4 starts on the next page]

Question 4.

Describe the factors that affect renal and biliary excretion of drugs. (20 marks)

End of Question 4

End of Paper PHA/2